CEUE203 - OOP25AIML034

PRACTICAL – 2

Aim:

1. Thermostat.java — Aim: To implement a smart thermostat using constructors, encapsulation, and static variables.

2. CinemaShow.java — Aim: To develop a cinema booking system using constructors, booking, cancellation, and static variables.

3. ParkingLot.java — Aim: To implement a parking lot system to manage vehicle parking, capacity, and revenue.

Code Screenshot:

```
public class Thermostat {

    private String location;

    private int temperature;

    private static final int MIN = 16;

    private static final int MAX = 30;

    private static int activeCount = 0;


    public Thermostat(String location, int startTemp) {

        this.location = location;
```

```
    if (startTemp >= MIN && startTemp <= MAX) {
        temperature = startTemp;
    } else {
        temperature = 22;
    }
    activeCount++;
}

public Thermostat(String location) {
    this(location, 22);
}

public void raise() {
    if (temperature < MAX) {
        temperature++;
    } else {
        System.out.println("Already at maximum (30)");
    }
}

public void lower() {
    if (temperature > MIN) {
```

```
        temperature--;
    } else {
        System.out.println("Already at minimum (16)");
    }
}

public int getTemperature() {
    return temperature;
}

public static int getActiveCount() {
    return activeCount;
}

public static void main(String[] args) {
    Thermostat t1 = new Thermostat("Room 1", 22);
    Thermostat t2 = new Thermostat("Room 2");

    for (int i = 0; i < 10; i++) {
        t1.raise();
        System.out.println("Temperature: " +
                           t1.getTemperature());
```

```
        }

        for (int i = 0; i < 20; i++) {
            t1.lower();
            System.out.println("Temperature: " +
                    t1.getTemperature());
        }

        System.out.println("Active count: " +
                Thermostat.getActiveCount());
    }
}
```

2.

```
public class CinemaShow {
    private String title;
    private int seatsAvailable;
    private final int capacity;
    private static int totalBooked = 0;

    public CinemaShow(String title, int capacity) {
        this.title = title;
```

```
    this.capacity = capacity;
    this.seatsAvailable = capacity;
}

public CinemaShow(String title) {
    this(title, 100);
}

public boolean book(int n) {
    if (n > 0 && n <= seatsAvailable) {
        seatsAvailable -= n;
        totalBooked += n;
        return true;
    }
    return false;
}

public void cancel(int n) {
    if (n > 0) {
        seatsAvailable = Math.min(capacity,
                                  seatsAvailable + n);
    }
```

```
}

public int getSeatsAvailable() {
    return seatsAvailable;
}

public static int getTotalBooked() {
    return totalBooked;
}

public static void main(String[] args) {
    CinemaShow show = new CinemaShow("Movie", 50);

    System.out.println("Book 20: " + show.book(20));
    System.out.println("Seats: " + show.getSeatsAvailable());

    System.out.println("Book 40: " + show.book(40));
    System.out.println("Seats: " + show.getSeatsAvailable());

    System.out.println("Book 10: " + show.book(10));
    System.out.println("Seats: " + show.getSeatsAvailable());
```

```
        show.cancel(5);
        System.out.println("After cancel: " +
                        show.getSeatsAvailable());

        System.out.println("Total booked: " +
                        CinemaShow.getTotalBooked());
    }
}
```

3.

```
public class ParkingLot {
    private int twoWheelers = 0;
    private int fourWheelers = 0;

    private final int twoCap;
    private final int fourCap;

    private static long revenue = 0;

    public ParkingLot(int twoCap, int fourCap) {
        this.twoCap = twoCap;
        this.fourCap = fourCap;
```

```
}

public void park(String type) {
    if (type.equalsIgnoreCase("two")) {
        if (twoWheelers < twoCap) {
            twoWheelers++;
            revenue += 20;
            System.out.println("Two-wheeler parked");
        } else {
            System.out.println("Full");
        }
    } else if (type.equalsIgnoreCase("four")) {
        if (fourWheelers < fourCap) {
            fourWheelers++;
            revenue += 40;
            System.out.println("Four-wheeler parked");
        } else {
            System.out.println("Full");
        }
    } else {
        System.out.println("Invalid type");
    }
```

```
}

public void leave(String type) {
    if (type.equalsIgnoreCase("two")) {
        if (twoWheelers > 0) {
            twoWheelers--;
            System.out.println("Two-wheeler left");
        } else {
            System.out.println("No two-wheelers parked");
        }
    } else if (type.equalsIgnoreCase("four")) {
        if (fourWheelers > 0) {
            fourWheelers--;
            System.out.println("Four-wheeler left");
        } else {
            System.out.println("No four-wheelers parked");
        }
    } else {
        System.out.println("Invalid type");
    }
}
```

```
    public void display() {
        System.out.println("Two-wheelers: " + twoWheelers);
        System.out.println("Four-wheelers: " + fourWheelers);
        System.out.println("Revenue: " + revenue);
    }

    public static void main(String[] args) {
        ParkingLot lot = new ParkingLot(2, 1);

        lot.park("two");
        lot.park("two");
        lot.park("two");
        lot.park("four");
        lot.park("four");

        lot.leave("two");
        lot.park("two");

        lot.display();
    }
}
```

Output Screenshot:

```
Temperature: 23
Temperature: 24
Temperature: 25
Temperature: 26
Temperature: 27
Temperature: 28
Temperature: 29
Temperature: 30
Already at maximum (30)
Temperature: 30
Already at maximum (30)
Temperature: 30
Temperature: 29
Temperature: 28
Temperature: 27
Temperature: 26
Temperature: 25
Temperature: 24
Temperature: 23
Temperature: 22
Temperature: 21
Temperature: 20
Temperature: 19
Temperature: 18
Temperature: 17
Temperature: 16
Already at minimum (16)
Temperature: 16
Already at minimum (16)
Temperature: 16
Already at minimum (16)
Temperature: 16
```

Output

```
Book 20: true
Seats: 30
Book 40: false
Seats: 30
Book 10: true
Seats: 20
After cancel: 25
Total booked: 30
```

Output

```
Two-wheeler parked
Two-wheeler parked
Full
Four-wheeler parked
Full
Two-wheeler left
Two-wheeler parked
Two-wheelers: 2
Four-wheelers: 1
Revenue: 100
```